

Containers and Kubernetes TASK – 5

1.Create a simple Node.js Express app that displays 'Welcome to My Playlist' at the root route, then write a Dockerfile to containerize this app and build the image locally.

Ans :

```
mkdir playlist-app
```

```
cd playlist-app
```

```
cat > package.json <<'EOF'
```

```
{  
  "name": "playlist-app",  
  "version": "1.0.0",  
  "main": "app.js",  
  "scripts": {  
    "start": "node app.js"  
  },  
  "dependencies": {  
    "express": "^4.21.2"  
  }  
}
```

```
EOF
```

```
cat > app.js <<'EOF'
```

```
const express = require("express");
```

```
const app = express();
```

```
app.get("/", (req, res) => {  
  res.send("Welcome to My Playlist");  
});
```

```
app.listen(3000, () => {  
  console.log("Server running on port 3000");  
});
```

EOF

```
cat > Dockerfile <<'EOF'
```

```
FROM node:22-alpine
```

```
WORKDIR /app
```

```
COPY package*.json ./
```

```
RUN npm install
```

```
COPY app.js .
```

```
EXPOSE 3000
```

```
CMD ["npm", "start"]
```

EOF

```
npm install
```

```
docker build -t playlist-app:latest .
```

```
docker run --rm -p 3000:3000 playlist-app:latest
```

**2.Push your container image to Docker Hub using your own account and tag it as playlist-app:latest.

Hint: Use 'docker tag' and 'docker push' commands after logging in to Docker Hub.**

Ans :

```
docker login
```

```
docker tag playlist-app:latest  
YOUR_DOCKERHUB_USERNAME/playlist-app:latest
```

```
docker push YOUR_DOCKERHUB_USERNAME/playlist-app:latest
```

```
docker images | grep playlist-app
```

```
YOUR_DOCKERHUB_USERNAME/playlist-app:latest
```

3.Set up a GitHub Actions workflow (YAML file) in your repository to automatically build and push the Docker image to Docker Hub whenever you push to the main branch.

Ans :

name: Build and Push Docker Image

on:

push:

branches:

- main

jobs:

docker:

runs-on: ubuntu-latest

steps:

- name: Checkout repository

uses: actions/checkout@v4

- name: Log in to Docker Hub

uses: docker/login-action@v3

with:

username: \${ secrets.DOCKERHUB_USERNAME }

password: \${ secrets.DOCKERHUB_TOKEN }

- name: Build and push Docker image

uses: docker/build-push-action@v6

with:

```
context: .  
  
push: true  
  
tags: ${{ secrets.DOCKERHUB_USERNAME }}/playlist-  
app:latest
```

**4.Modify your GitHub Actions workflow to also deploy the updated container image to a Kubernetes cluster using kubectl apply, referencing a deployment YAML file in your repo.

Hint: Use the 'kubectl set image' command to update the deployment with the new image tag.**

Ans :

```
apiVersion: apps/v1  
kind: Deployment  
metadata:  
  name: playlist-app  
spec:  
  replicas: 2  
  selector:  
    matchLabels:  
      app: playlist-app  
  template:  
    metadata:  
      labels:  
        app: playlist-app  
    spec:
```

containers:

- name: playlist-app

image: YOUR_DOCKERHUB_USERNAME/playlist-app:latest

ports:

- containerPort: 3000

name: Build, Push and Deploy

on:

push:

branches:

- main

jobs:

build-push-deploy:

runs-on: ubuntu-latest

steps:

- name: Checkout repository

uses: actions/checkout@v4

- name: Log in to Docker Hub

uses: docker/login-action@v3

with:

username: \${{ secrets.DOCKERHUB_USERNAME }}

password: \${{ secrets.DOCKERHUB_TOKEN }}

- name: Build and push Docker image

uses: docker/build-push-action@v6

with:

context: .

push: true

tags: \${{ secrets.DOCKERHUB_USERNAME }}/playlist-
app:\${{ github.sha }}

- name: Set up kubectl

uses: azure/setup-kubectl@v4

with:

version: latest

- name: Configure Kubernetes

uses: azure/k8s-set-context@v4

with:

method: kubeconfig

kubeconfig: \${{ secrets.KUBE_CONFIG }}

- name: Apply Kubernetes deployment

run: kubectl apply -f k8s/deployment.yaml

- name: Update deployment image

run: |

```
kubectl set image deployment/playlist-app \
    playlist-app=${{ secrets.DOCKERHUB_USERNAME
}}/playlist-app:${{ github.sha }}
```

- name: Verify deployment

run: kubectl rollout status deployment/playlist-app

DOCKERHUB_USERNAME

DOCKERHUB_TOKEN

KUBE_CONFIG

git add k8s/deployment.yaml .github/workflows/docker-publish.yml

git commit -m "Add Kubernetes deployment to CI/CD"

git push origin main

5. Use ChatGPT or GitHub Copilot to review your CI/CD workflow YAML and suggest at least two improvements for reliability or security. List the suggestions and explain how you would implement them.

Ans :

1. Create the Kubernetes deployment file

mkdir -p k8s

cat > k8s/deployment.yaml <<'EOF'

apiVersion: apps/v1

```
kind: Deployment
metadata:
  name: playlist-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: playlist-app
  template:
    metadata:
      labels:
        app: playlist-app
    spec:
      containers:
        - name: playlist-app
          image: YOUR_DOCKERHUB_USERNAME/playlist-
app:latest
          ports:
            - containerPort: 3000
EOF
```

2. Create the improved GitHub Actions workflow

```
mkdir -p .github/workflows
```

```
cat > .github/workflows/docker-publish.yml <<'EOF'
```

```
name: Build, Push and Deploy
```

```
on:
```

```
  push:
```

```
    branches:
```

```
      - main
```

```
permissions:
```

```
  contents: read
```

```
jobs:
```

```
  build-push-deploy:
```

```
    runs-on: ubuntu-latest
```

```
    steps:
```

```
      - name: Checkout repository
```

```
        uses: actions/checkout@v4
```

```
      - name: Log in to Docker Hub
```

```
        uses: docker/login-action@v3
```

```
        with:
```

```
          username: ${ secrets.DOCKERHUB_USERNAME }
```

```
          password: ${ secrets.DOCKERHUB_TOKEN }
```

```
# Use the Git commit SHA as a unique image tag
- name: Build and push Docker image
  uses: docker/build-push-action@v6
  with:
    context: .
    push: true
    tags: ${{ secrets.DOCKERHUB_USERNAME }}/playlist-
app:${{ github.sha }}
```

```
- name: Set up kubectl
  uses: azure/setup-kubectl@v4
  with:
    version: latest
```

```
- name: Configure Kubernetes
  uses: azure/k8s-set-context@v4
  with:
    method: kubeconfig
    kubeconfig: ${{ secrets.KUBE_CONFIG }}
```

```
- name: Apply Kubernetes deployment
  run: kubectl apply -f k8s/deployment.yaml
```

```
# Deploy the exact image produced by this commit
- name: Update deployment image
  run: |
    kubectl set image deployment/playlist-app \
      playlist-app=${{ secrets.DOCKERHUB_USERNAME
    }}/playlist-app:${{ github.sha }}
```

```
- name: Verify deployment
  run: kubectl rollout status deployment/playlist-app
EOF
```

3. Replace the placeholder username in the deployment file

with your actual Docker Hub username

```
sed -i
's/YOUR_DOCKERHUB_USERNAME/YOUR_ACTUAL_DOCKERHUB_USERNAME/g' k8s/deployment.yaml
```

4. Add these GitHub Secrets in:

GitHub repository → Settings → Secrets and variables → Actions

#

DOCKERHUB_USERNAME = your Docker Hub username

DOCKERHUB_TOKEN = your Docker Hub access token

KUBE_CONFIG = your Kubernetes kubeconfig

#

Do NOT put passwords, tokens, or kubeconfig directly in the YAML.

5. Commit the improved workflow

```
git add k8s/deployment.yaml .github/workflows/docker-publish.yml
```

```
git commit -m "Improve CI/CD reliability and security"
```

```
git push origin main
```

6. After pushing, check the workflow

GitHub → Actions → Build, Push and Deploy

#

The workflow now:

- Runs only on main

- Uses read-only repository permissions

- Builds the Docker image

- Pushes a unique image tagged with the Git commit SHA

- Applies the Kubernetes deployment

- Updates the deployment to the exact image built

- Waits for the Kubernetes rollout to complete